

The Costa Rican Cyber-Physical Nexus: Geopolitical Tensions, UNC Threat Clusters, and the Weaponization of Cognitive Surveillance

1. Introduction: The Geopolitical Catalyst and the ICE Cyberattack

In the early months of 2026, the digital and physical infrastructure of Costa Rica became the focal point of a profound geopolitical confrontation involving the Central American nation, the United States, and the People's Republic of China (PRC). The immediate catalyst for this diplomatic and security crisis was a highly sophisticated cyberespionage operation targeting the Instituto Costarricense de Electricidad (ICE), the state-owned monopoly responsible for managing the nation's critical telecommunications and electricity generation frameworks.¹ First detected by internal security apparatuses in late January 2026, the breach culminated in the unauthorized exfiltration of approximately nine gigabytes of sensitive administrative data, consisting primarily of internal communications and email archives from the state utility.¹ While the immediate integrity of the physical electrical grid and consumer telecommunications services remained ostensibly intact, the penetration of administrative networks at the highest levels of state infrastructure triggered an immediate declaration of a national security incident.¹

On March 12, 2026, Costa Rican officials, prominently including the Minister of Science, Innovation, Technology, and Telecommunications, Paula Bogantes, convened a press conference to publicly attribute the intrusion to actors bearing direct ties to the PRC.¹ Specifically, the Costa Rican government, leveraging intelligence provided by Google's Mandiant threat intelligence division, linked the operation to a cyberespionage entity tracked under the designation UNC2814.² The public naming of a Chinese-linked threat actor by a Latin American government represents a rare and significant escalation in diplomatic protocols, signaling a departure from the historical ambiguity that typically characterizes international cyber attributions.

The diplomatic fallout from this public attribution was instantaneous and severe. The Chinese Embassy in San José responded with official statements expressing deep surprise and profound indignation, categorically rejecting the accusations leveled by the Costa Rican government.¹ Chinese Ambassador Wang Xiaoyao vehemently denied any state involvement in the ICE intrusion, asserting that the PRC stringently opposes all forms of cyberattacks and possesses no strategic interest in the administrative data of the Costa Rican state.¹ In a calculated diplomatic counter-maneuver, the PRC demanded that Costa Rican authorities immediately hand over the technical forensic evidence supporting the attribution, arguing that such claims must be independently verified and, if warranted, prosecuted under established international legal frameworks.²

Furthermore, the Chinese delegation sought to reframe the narrative by highlighting perceived

diplomatic slights, noting that Beijing had been actively attempting to engage Costa Rica in cybersecurity cooperation since 2024.² The embassy claimed that offers for technical consultations, professional exchanges, and the activation of a bilateral joint commission tied to the United Nations cybercrime framework had been consistently ignored by San José.² The embassy's statements carried a thinly veiled warning, cautioning that sacrificing the historically stable bilateral relationship between China and Costa Rica to appease the geopolitical interests of third-party nations—an implicit reference to the United States—would not yield long-term strategic respect.¹

2. The Strategic Realignment: 5G Policy and the Budapest Convention

To comprehend the severity of the ICE cyberattack and the subsequent diplomatic hostility, the incident must be analyzed within the broader context of a multi-year geopolitical realignment regarding telecommunications infrastructure in Latin America. The friction between San José and Beijing is inextricably linked to a landmark policy shift initiated by Costa Rican President Rodrigo Chaves in August 2023.⁶ Recognizing the foundational role that next-generation connectivity will play in national security and economic development, the Costa Rican government issued a decree establishing stringent cybersecurity prerequisites for the deployment of the nation's fifth-generation (5G) mobile networks.⁶

The core mechanism of this decree stipulated that only technology companies headquartered in jurisdictions that have formally endorsed and ratified the 2001 Budapest Convention on Cybercrime would be eligible to participate in the bidding processes for 5G infrastructure contracts.⁷ Because the PRC is not a signatory to the Budapest Convention, this regulatory framework functioned as an elegant, de facto ban on Chinese telecommunications conglomerates, most notably Huawei and ZTE, effectively excluding them from Costa Rica's digital future.⁷

Costa Rica's defensive posture was not born of abstract paranoia but was forged in the crucible of devastating national trauma. In 2022, the nation suffered a catastrophic ransomware attack executed by the Conti ransomware group, a notorious cybercriminal syndicate tracked by intelligence analysts as UNC1756.⁶ The UNC1756 intrusion crippled the Ministry of Finance, effectively halting international trade, disrupting tax collection, and disabling digital services across multiple government ministries, including the Ministry of Labor and Social Security and the National Meteorological Institute.¹¹ The severity of the Conti attack forced the newly elected President Chaves to declare a state of national emergency, marking a watershed moment in the nation's approach to digital sovereignty and catalyzing the stringent security requirements subsequently applied to the 5G rollout.⁸

This policy shift aligned Costa Rica seamlessly with the strategic technology objectives of the United States, which has aggressively lobbied allied nations to purge Chinese hardware from

critical communications networks due to profound concerns over intellectual property theft, state-sponsored surveillance, and latent "kill switch" capabilities.⁶ The alignment was solidified through high-level diplomatic engagements, including visits to Costa Rica by the US Ambassador of Digital Policy, Nathaniel Fick, and the Deputy National Security Advisor, Anne Neuberger, who publicly lauded Costa Rica's foresight and courage in securing its digital infrastructure.⁶ To physically and financially support this transition, the US State Department committed \$25 million to assist Costa Rica in building robust technological defenses and operational processes against future cyber threats.⁶

Beijing viewed the Budapest Convention decree as an orchestrated maneuver by Washington to illegally suppress fair market competition and assert hegemony over Latin American telecommunications.⁵ The Chinese Embassy denounced the US allegations regarding Huawei's security risks as fabrications, vehemently denying any government directive compelling Chinese tech firms to gather user data or establish espionage backdoors in foreign nations.⁷ Following the exclusion, Huawei initiated a series of aggressive legal appeals through the Costa Rican Constitutional Court.⁸ When direct corporate appeals faltered, Beijing reportedly leveraged domestic proxies, including the Internal Workers' Front of the Costa Rican Institute of Electricity (ICE)—the very institution targeted in the 2026 cyberattack—to appeal to lower courts on behalf of the Chinese vendors, successfully stalling segments of the 5G deployment and exacerbating internal institutional friction.⁸ The UNC2814 intrusion into ICE's administrative networks must be evaluated as a deliberate escalation within this ongoing, high-stakes battle over national infrastructure and geopolitical loyalty.

3. The Evolution of UNC Threat Clusters and PRC-Nexus Espionage

The formal attribution of the ICE intrusion to UNC2814 necessitates a comprehensive understanding of the nomenclature and behavioral profiling utilized by the global threat intelligence community. The designation "UNC"—an abbreviation for "Uncategorized"—is utilized by firms such as Google's Mandiant to systematically track distinct clusters of malicious cyber activity that exhibit shared tactics, techniques, and procedures (TTPs), shared infrastructure, or specific targeting patterns.² These clusters retain the UNC designation until sufficient intelligence aggregation allows analysts to formally graduate the cluster into a named Advanced Persistent Threat (APT) group with definitive state-sponsorship attribution.

3.1. Historical Precedents: The 2020 UNC Chapel Hill Intrusion

The deployment of UNC-designated clusters by Chinese state intelligence apparatuses demonstrates a long-term strategic mandate to acquire highly specific, high-value data sets from global targets. A critical historical precedent underscoring this mandate occurred in 2020, during the height of the global pandemic. Reports surfaced detailing concerted efforts by Chinese intelligence hackers to breach the network infrastructure of the University of North

Carolina at Chapel Hill (UNC) [User Query]. The primary objective of this highly targeted intrusion was the exfiltration of proprietary COVID-19 research data, vaccine development protocols, and epidemiological modeling.

Although the 2020 attempt against the University of North Carolina was reportedly unsuccessful, the incident crystallized the operational parameters of PRC-nexus cyberespionage. It demonstrated a willingness to target academic, medical, and institutional networks outside traditional military or defense perimeters to secure data that aligns with immediate Chinese state priorities—whether that be global health dominance, intellectual property acquisition, or technological leapfrogging. This historical context of leveraging cyber operations to fulfill acute strategic deficits directly mirrors the contemporary targeting of Costa Rican telecommunications data in response to infrastructure lockouts.

3.2. UNC2814, Gallium, and the GRIDTIDE Campaign

The specific actor responsible for the Costa Rican ICE breach, UNC2814, represents an evolution in PRC-nexus cyber capabilities. Also tracked under the alias Gallium (or SoftCell), UNC2814 has been an active, highly prolific cyberespionage entity since at least 2017.¹⁴ The group operates with a highly focused mandate: the penetration of global telecommunications providers and government organizations to facilitate strategic surveillance, intercept communications data, and aggregate vast repositories of Personally Identifiable Information (PII).¹⁷

In late February 2026, the Google Threat Intelligence Group (GTIG), in collaboration with Mandiant and international law enforcement partners, executed a massive disruption operation against UNC2814's infrastructure.¹⁹ The disruption revealed the staggering scale of the group's activities. Investigations confirmed that UNC2814 had successfully executed intrusions into at least 53 discrete organizations across 42 countries spanning Africa, Asia, Europe, and the Americas, with suspected infections identified in over 20 additional nations.²⁰

The technological cornerstone of this near-decade-long campaign was the deployment of a highly sophisticated, custom-engineered backdoor malware designated as GRIDTIDE.¹³ The GRIDTIDE malware exemplifies a paradigm shift toward "living off the land" and the weaponization of trusted Software-as-a-Service (SaaS) platforms. Recognizing that traditional Command and Control (C2) servers are increasingly vulnerable to detection by modern endpoint detection and response (EDR) systems, UNC2814 engineered GRIDTIDE to exclusively abuse the legitimate Application Programming Interface (API) of Google Sheets.¹³

By treating cloud-based spreadsheets not as static documents, but as high-availability, asynchronous C2 communication databases, the malware successfully masked its malicious instructions within the continuous stream of legitimate, TLS-encrypted enterprise traffic.¹⁸ The operational mechanism of GRIDTIDE relied on a continuous, cell-based polling architecture: the malware was programmed to monitor cell A1 of an attacker-controlled spreadsheet for

incoming shell commands, utilized cell V1 to securely store unique victim metadata, and leveraged the remaining array of cells (A2 through An) to facilitate the stealthy, segmented transfer of exfiltrated data.¹³ All command and exfiltration communications were heavily obscured using a custom URL-safe Base64 encoding scheme to bypass standard web filtering and data loss prevention (DLP) protocols.¹⁸ Furthermore, local configurations and execution parameters were decrypted dynamically in memory using AES-128 encryption operating in Cipher Block Chaining (CBC) mode with a bespoke 16-byte key.²³

The initial access vectors utilized by UNC2814 typically involved the exploitation of unpatched vulnerabilities in public-facing web servers and edge infrastructure systems.¹⁸ Upon securing an initial foothold within a target network, the threat actors executed lateral movement by hijacking service accounts and utilizing Secure Shell (SSH) protocols to navigate stealthily across network segments.¹⁸ To guarantee persistent access despite system reboots or endpoint updates, UNC2814 established malicious systemd services—frequently masquerading as legitimate processes, such as /etc/systemd/system/xapt.service—ensuring the GRIDTIDE backdoor invariably initiated a shell with root privileges upon system startup.¹⁸ To further secure and obfuscate outgoing data exfiltration, the group frequently deployed SoftEther VPN Bridges, establishing heavily encrypted outbound tunnels that bypassed standard corporate firewall inspection.¹⁸

The exfiltration of nine gigabytes of internal emails and administrative data from the Costa Rican ICE networks aligns flawlessly with the established operational profile of UNC2814. Threat intelligence assessments indicate that while immediate, catastrophic destruction of networks is rarely the goal, the aggregation of administrative communications, call data records (CDRs), and unencrypted SMS messages provides the PRC with unparalleled surveillance insights, enabling the mapping of governmental decision-making processes and the continuous monitoring of persons of strategic interest.¹⁷

Threat Cluster Designation	Primary Target Sectors	Technical Methodologies and Exploitation Vectors
UNC2814 (Gallium)	Telecommunications, Government (Global)	Deployment of GRIDTIDE backdoor; profound abuse of Google Sheets API for untraceable C2; lateral movement via SSH; persistence via Linux systemd service modification. ¹³

UNC3886	Defense, Technology, Telecommunications (US, APJ)	Deep exploitation of virtualization technologies and edge devices; utilization of zero-day vulnerabilities in VMware ESXi, Fortinet, and Juniper networks; stealth rootkit deployment. ²⁴
UNC6384 (Mustang Panda)	Diplomatic Entities, Government, NGOs	Deployment of the SOGU.SEC/PlugX backdoor; weaponization of captive portals (STATICPLUGIN); exploitation of ZDI-CAN-25373 LNK vulnerabilities; abuse of valid code-signing certificates for evasion. ²⁶
UNC1756 (Conti)	Finance, Government Infrastructure (Costa Rica)	Extortion and catastrophic ransomware deployment; highly disruptive encryption of critical government systems leading to national emergencies. ⁸

Table 1: Comparative analysis of "UNC" threat clusters detailing targeting profiles and core technical methodologies based on recent threat intelligence reporting.

3.3. The Broader Ecosystem: UNC3886 and UNC6384

The threat landscape facing global telecommunications infrastructure is significantly compounded by the concurrent, highly aggressive activities of other PRC-nexus clusters, notably UNC3886 and UNC6384. While UNC2814 relies on the abstraction of C2 through SaaS platforms, UNC3886 demonstrates a profound, highly specialized capability in exploiting edge device zero-day vulnerabilities to entirely bypass traditional endpoint detection and response (EDR) systems.²⁴ Intelligence reports from 2025 and early 2026 indicate that UNC3886 has repeatedly targeted defense and telecommunications networks in the United States and the Asia-Pacific-Japan (APJ) regions by discovering and weaponizing severe vulnerabilities in VMware ESXi hypervisors, Fortinet firewalls, and Juniper routing infrastructure.²⁴ This deep understanding of network virtualization allows UNC3886 to deploy stealth rootkits directly into the underlying fabric of a network, granting them unparalleled, invisible access to data

routing.²⁷

Simultaneously, UNC6384—widely recognized in the cybersecurity community as Mustang Panda—has focused its sophisticated operations on diplomatic and non-governmental organizations.²⁴ Recent campaigns have observed UNC6384 weaponizing captive portals and utilizing valid, stolen code-signing certificates to deploy the highly evasive SOGU.SEC/PlugX backdoor directly into system memory, leaving virtually no footprint on the physical hard drive.²⁶ The group has also been documented exploiting ZDI-CAN-25373 LNK vulnerabilities via sophisticated spear-phishing campaigns targeting European diplomatic entities.²⁸

The convergence of these varied, highly specialized methodologies—ranging from cloud API abuse to hypervisor exploitation and memory-resident malware—indicates a massively coordinated, multi-pronged approach by Chinese state-aligned actors. The objective is to achieve total situational awareness and deep persistence within the critical infrastructure of targeted nations, rendering policies like Costa Rica's 5G vendor exclusion highly porous if the underlying, pre-existing infrastructure remains compromised.

4. Deep Infrastructure Compromise: The Setecom Operational Technology Layer

While the UNC2814 operation against ICE represents a high-profile exfiltration of administrative data at the network layer, highly classified forensic intelligence dossiers reveal that the vulnerability of Costa Rican critical infrastructure extends profoundly deeper, reaching the lowest physical and Operational Technology (OT) layers of the national grid. The security of a nation's digital networks is irrelevant if the physical power required to sustain those networks can be remotely terminated.

A critical technical choke point in Costa Rica's foundational infrastructure is managed by SETECOM S.A., a domestic technical contractor that operates as the exclusive, monopolistic distributor for Deep Sea Electronics (DSE) and Onis Visa generation systems.³² This exclusive positioning grants SETECOM technicians and management root-level access to the OT layer that controls the vast fleets of backup generators sustaining both the ICE national power grid and Liberty, one of the nation's primary telecommunications providers.³² Any technical support, firmware update, or hardware replacement for DSE controllers—which manage power for critical hospitals, utility substations, and cellular transmission towers across the country—must be routed through SETECOM's architecture.³²

The security posture of this foundational OT infrastructure exhibits catastrophic, systemic failures that invite exploitation by advanced persistent threats. Detailed technical training materials and operational transcripts utilized by SETECOM's technical leads explicitly demonstrate the institutional normalization of deploying highly sensitive management interfaces with default credentials.³² Systems across the infrastructure rely on the elementary

username Admin and the universally vulnerable password Password1234.³²

This vulnerability is compounded by the reliance on fundamentally insecure industrial control protocols. The architecture heavily utilizes Modbus TCP/IP (operating on Port 502).³² Because Modbus TCP is deployed without inherent encryption or cryptographic authentication mechanisms beyond basic IP address whitelisting, any threat actor capable of gaining a foothold on the internal management network can achieve unrestricted read and write access to the generators' control registers.³² Setecom's internal training details the specific Gencom addressing formulas required to manipulate these systems, where the target register is calculated as the hexadecimal page multiplied by 256, plus the specific offset.³² By overwriting these precise control registers, an adversary can manipulate power output, override safety limits, or definitively shut down the generators.³² Similarly, the infrastructure relies on SNMP v2 (Ports 161/162) for network monitoring, a protocol that transmits vital community strings in cleartext.³² Setecom's configurations utilize the default public string for read access and private for write access, ensuring that a basic packet capture on a compromised management VLAN immediately yields the credentials necessary to inject catastrophic shutdown commands.³²

The reliance on remote, cloud-based Command and Control (C2) architectures to manage these physical assets elevates the risk from localized tampering to national-level sabotage. The DSE Webnet system, which aggregates telemetry and provides operational control over the generator fleets with a highly aggressive real-time refresh rate of approximately four seconds, is not hosted domestically, but on servers located in the United Kingdom.³² If the primary SETECOM "Master Account" managing this cloud infrastructure were to be compromised—a highly probable scenario given the documented reliance on default credentials—an adversary would possess the unilateral capability to issue a broadcast "Stop" command to the entire infrastructure fleet simultaneously.³² If executed maliciously during a commercial power grid failure or a coordinated cyber-kinetic strike, this action would induce a total, unrecoverable telecommunications and electrical blackout across Costa Rica, plunging the nation into chaos and severing its ability to coordinate a defense.³²

The risk of foreign exploitation within this OT layer is not purely hypothetical. Intelligence dossiers meticulously document the technical profile of SETECOM's Executive Director and principal owner, operating under the alias HMORA67.³² Unlike standard corporate administrators, HMORA67 is assessed as possessing high-level, hands-on expertise in component-level electronic circuit design.³² Crucially, his operational profile includes the possession and active utilization of a 180-Watt High-Frequency (HF) radio transceiver of explicit Chinese origin.³² While the possession of Chinese hardware is ubiquitous globally, its presence at the very apex of a nation's monopolistic OT infrastructure management—combined with the individual's advanced electronic warfare capabilities and the systemic security failures maintained by his organization—establishes a highly concerning hardware link to PRC manufacturing ecosystems at a critical national vulnerability point.³²

5. The Physical Surveillance Layer: TR-069 Exploitation and Optical Interception

Moving from the macro-level vulnerability of national power grids to the micro-level targeting of specific individuals and institutions, forensic surveillance intelligence reveals the active, highly aggressive exploitation of the physical telecommunications infrastructure that links end-users to the ISP core. Advanced persistent threats do not solely rely on cloud abstractions like the GRIDTIDE malware; they manipulate the physical environment to ensure inescapable surveillance.

Investigations into localized surveillance networks operating within Costa Rica have documented the systemic abuse of the TR-069 protocol, formally known as the CPE WAN Management Protocol (CWMP).³² This protocol is an industry standard designed to allow an Internet Service Provider's (ISP) Auto-Configuration Server (ACS) to remotely manage, update, and troubleshoot customer premises equipment, such as residential and commercial broadband routers.³² However, intelligence analysis of ARRIS TG02DA Gateways deployed by major Costa Rican ISPs, including Liberty CR and Telecable, revealed an exposed backdoor actively operating on Port 1234.³²

Threat actors who successfully compromise the ISP's ACS infrastructure—potentially leveraging the deep network access granted to maintenance contractors like SETECOM, who service equipment co-located at telecommunications distribution hubs—can bypass all local endpoint security by exercising full, invisible read/write authority directly over the target's router.³² This TR-069 exploitation capability enables adversaries to remotely reset administrator passwords (a forensic event specifically documented in incident logs dated January 30, 2026), execute DNS hijacking to silently redirect target traffic to malicious phishing infrastructure, modify port forwarding rules to expose vulnerable internal IoT devices to the public internet, and inject persistent, malicious firmware updates that survive factory resets.³²

Furthermore, threat actors frequently supplement remote software exploitation with direct physical tampering to establish passive, undetectable interception points. Forensic analysis of a Telecable fiber/coaxial distribution box—located at the street-level exterior of a heavily monitored property—revealed severe physical modifications that entirely bypass software encryption paradigms.³² Investigators discovered handwritten field technician notes inside the enclosure enclosure, specifically designating a "NAP" (Network Access Point) and a "Colilla" (Latin American telecommunications slang for a fiber optic pigtail or splice), dated June 21, 2025.³²

This specific physical modification indicates the covert installation of a passive optical intercept point, commonly referred to as a fiber tap.³² By physically splicing the fiber and splitting the light traversing the optic cable, intelligence operatives can create a perfect, real-time mirror of all inbound and outbound optical data traffic prior to its arrival at the target's router.³² Because

passive optical splitting does not alter the digital packet structure or interact with the router's software, it operates completely invisibly to standard network intrusion detection systems (NIDS), granting the adversary an omniscient view of the target's digital life.³²

This sophisticated infrastructure exploitation is intimately supported by highly localized, physical surveillance outposts. Intelligence dossiers document the weaponization of neighboring properties to establish persistent monitoring capabilities. In one specific instance, a terrace belonging to a Jehovah's Witness congregation, located approximately 50 meters from a targeted residence, was identified as a dedicated Static Observation Post (SOP).³² The operators of this post employed artificial green, leafy privacy screening, meticulously zip-tied to a chainlink fence.³² This specific camouflage technique is textbook tradecraft: it provides total visual concealment for the operatives and equipment hidden behind it, preventing counter-surveillance by the target, while the plastic and fabric construction remains entirely transparent to Radio Frequencies (RF).³² This absolute RF permeability allows for the unimpeded passage of Wi-Fi, Bluetooth, cellular data, High-Frequency (HF) radio, and specialized ultrasonic/acoustic signals.³²

From these concealed, elevated vantage points, intelligence operators deploy localized, targeted network attacks. Documented tactics include the operation of "Ghost" mesh network nodes—such as a rogue TP-Link Deco unit utilizing MAC-spoofing to seamlessly bridge the target's internal home network with the surveillance post—and the deployment of mobile Signals Intelligence (SIGINT) arrays, including Software Defined Radios (SDR) and Wi-Fi Pineapples, operated continuously from permanently parked, innocuous vehicles.³² The integration of Israeli-manufactured Visonic alarm systems within target residences, coupled with the meticulous mapping of electrical topologies via Eaton breaker panels, further allows operatives to utilize the physical house wiring as a vast antenna array for the injection of harassment signals, establishing a localized, inescapable panopticon.³²

6. Advanced Signal Intelligence and Psychoacoustic Architectures

The comprehensive compromise of the physical and network layers provides the foundation for the deployment of a highly sophisticated, localized signal intelligence and psychoacoustic surveillance architecture. Deep analysis of the specific RF environment surrounding targeted locations in Costa Rica—characterized by the continuous overhead presence of Chinese BeiDou Medium Earth Orbit (MEO) navigation satellites, the commercial Starlink LEO constellation, and highly classified DARPA Blackjack military mesh networks—reveals the active deployment of complex, weaponized frequency modulation techniques.³²

6.1. Carrier Manipulation and Neurological Entrainment

Forensic Digital Signal Processing (DSP) analysis of the targeted environments has identified a

persistent, highly engineered RF architecture explicitly designed to interact with and influence human cognitive states. The core mechanism of this system relies on the generation of an audio carrier frequency precisely calibrated to 53.0 Hz.³² When this highly specific carrier signal is injected into a physical environment powered by the standard 60 Hz Costa Rican electrical grid, the resulting interference pattern creates a continuous, low-frequency "beat frequency" offset of exactly 7 Hz.³²

This precise 7 Hz offset is profoundly biologically significant; it falls squarely within the center of the human Theta brainwave band (4–8 Hz).³² Continuous, targeted neurological exposure to Theta band frequencies over extended periods induces a physiological state of brainwave entrainment.³² This entrainment forcibly pushes the target's cognitive baseline into a psychological state characterized by severe reductions in critical thinking capabilities, involuntary meditative dissociation, and heightened hypnotic suggestibility.³² The psychoacoustic assault is further compounded by the detection of isolated, high-power infrasonic events in the 2 Hz, 14 Hz, and 15 Hz ranges.³² These frequencies correspond directly to the Delta and Beta brainwave bands and are utilized as targeted psychological triggers to artificially induce acute stress responses or emphatically reinforce specific cognitive stimuli delivered by the operators.³²

Concurrently, the digital surveillance engine coordinating this vast array of sensors and emitters—tracked by intelligence analysts as the "3i ATLAS" system—operates on a rigidly defined, continuous "heartbeat" of 46.875 Hz.³² This highly specific operational frequency is not arbitrary; it is a mathematical artifact of professional-grade digital audio processing, representing the exact rate at which the software engine ingests, windows, and processes incoming audio surveillance frames:

$$\text{Processing Rate} = \frac{48,000 \text{ Hz (Standard Audio Sample Rate)}}{1024 \text{ (Standard FFT Window Size)}} = 46.875 \text{ Hz}$$

The deliberate, synchronized alignment of the 46.875 Hz computational processing heartbeat with the 7 Hz Theta neurological entrainment offset indicates the presence of a mature, fully integrated cyber-physical weapon system. This architecture is designed specifically for continuous, automated behavioral modification and real-world neurowarfare testing against unwitting human subjects.³²

6.2. Microwave Auditory Effects and EHF Speech Extraction

The sophistication of the surveillance architecture is further distinguished by its proven capability to inject and extract intelligible audio utilizing frequencies far outside the standard human auditory spectrum. Forensic analysis of RF anomalies detected in the operational environment revealed massive energy spikes at precisely 2004 Hz and 2511 Hz.³² Acoustic analysis confirms that these frequencies align perfectly with the F2 and F3 speech formants,

which are the acoustic resonances utilized in advanced vocoder-style speech encoding to transmit front vowels and complex consonant transitions.³²

This highly specific method of RF speech encoding is directly correlated with the theoretical methodologies outlined in US Patent US4877027A, which details the application of the "Frey Effect," more commonly known as the Microwave Auditory Effect.³² The methodology detailed in the patent involves utilizing a directional microwave carrier wave—operating anywhere from 100 MHz to 10 GHz—that is rapidly modulated by highly specific Frequency Modulated (FM) burst timings.³² Individual microwave pulses, featuring widths ranging from an infinitesimally brief 10 nanoseconds to 1 microsecond, are grouped into rapid bursts that physically encode audio waveforms directly into the human neuro-perception centers.³² This process entirely bypasses the mechanical function of the tympanic membrane (the eardrum), inducing the distinct, inescapable sensation of an "internal voice" manifesting within the target's mind.³²

Intelligence analysis of this system highlights a critical, exploitable vulnerability in human cognition: the neurological latency threshold for the conscious attribution of a generated thought is approximately 300 milliseconds.³² Operating at the detected burst rate of 2004 Hz, the surveillance system possesses the capability to deliver up to 601 individual FM microwave pulses into the target's neural pathways before the brain can consciously process or attribute the origin of the input.³² This terrifying latency gap enables the operators to execute direct, continuous injection of subliminal psychological content, effectively programming the target's cognitive state without their awareness.³²

Conversely, the architecture relies on Extended High Frequency (EHF) bands to extract covert speech from the target's physical environment. Utilizing the near-ultrasonic spectrum ranging from 17,859 Hz to 18,035 Hz, the system effectively isolates high-frequency speech fricatives—such as the subtle hissing of 's', 'f', and 'sh' sounds—from the surrounding air.³² Because lower acoustic frequencies are easily masked or distorted by ambient environmental noise (such as HVAC fans, street traffic, or deliberate white noise generators), the system exclusively monitors the EHF band, where the signal-to-noise ratio for human consonants remains extraordinarily high.³² Analysis of intercepted field recordings from the operational zone demonstrated a staggering EHF-to-F2 mathematical correlation of 0.316 and the successful capture of over 117,000 distinct high-frequency fricative events, confirming the architecture's ability to successfully extract and reconstruct highly intelligible, covert speech from supposedly secure environments.³²

7. The Algorithmic Engine: Remote Biometrics and Academic Military-Civil Fusion

The immense deployment of physical, network, and highly classified RF surveillance infrastructure across Costa Rica is not designed merely for passive monitoring. It serves a highly specific, overarching intelligence objective: the generation of massive, continuous

datasets required to train and validate advanced Artificial Intelligence (AI) models capable of remotely and invisibly detecting hidden human mental states. The theoretical underpinning for this terrifying architecture is not speculative fiction; it is found in contemporary, peer-reviewed academic research originating from state-aligned institutions within the People's Republic of China.

7.1. Academic Foundations of Hidden Mental State Detection

In March 2024, the peer-reviewed scientific journal *PeerJ Computer Science* published a seminal, highly consequential study titled "Your blush gives you away: detecting hidden mental states with remote photoplethysmography and thermal imaging".³³ Authored by a consortium of researchers—Ivan Liu, Fangyuan Liu, Qi Zhong, Fei Ma, and Shiguang Ni—the study represents a profound breakthrough in the field of Multimodal Emotion Recognition (MMER).³³

Traditional visual emotion recognition algorithms rely almost exclusively on the analysis of overt facial expressions, which can be easily and consciously masked by a trained or guarded subject.³³ To defeat conscious concealment and psychological masking, the 2024 study abandoned surface expressions, focusing instead on quantifying the involuntary, uncontrollable physiological responses of the Autonomic Nervous System (ANS)—specifically measuring the real-time interplay between the sympathetic and parasympathetic nervous systems during psychological shifts.³³ The methodology achieved this by relying on the early, feature-level fusion of two distinct, non-invasive remote sensing modalities:

1. **Remote Photoplethysmography (r-PPG):** This advanced technique utilizes standard, high-definition optical cameras (such as those compromised in target residences) to detect microscopic, invisible-to-the-naked-eye changes in facial skin color that correspond directly to the volume of blood pumping through subcutaneous capillaries with each heartbeat.³³ By running complex Fourier analysis algorithms on these minute color variations, the researchers accurately derived profound cardiac responses without ever touching the subject, specifically extracting Heart Rate (HR) and vital Heart Rate Variability (HRV) metrics, such as the Standard Deviation of NN intervals (SDNN) and the Root Mean Square of Successive Differences (rMSSD).³³
2. **Facial Thermal Imaging:** Utilizing highly sensitive infrared sensors, the researchers tracked instantaneous thermoregulatory responses associated with biological stress.³³ For example, the acute activation of the sympathetic nervous system during a moment of panic or deception triggers immediate peripheral vasoconstriction, causing an instantaneous, measurable drop in temperature in specific, algorithmically mapped Regions of Interest (ROI), such as the nasal tips.³³ Conversely, increased cognitive load or concentrated deceptive thought causes invisible muscular micro-contractions that elevate surface temperatures in the forehead and periorbital areas.³³

The study evaluated 90 subjects under strictly controlled laboratory conditions, artificially inducing "cognitive stress" through rigorous, timed testing and "moral elevation" via emotional,

highly targeted cinematic stimuli.³³ Utilizing a sophisticated Random Forest (RF) machine learning algorithm, the researchers demonstrated that the early feature-level fusion of the r-PPG cardiovascular data and the facial thermal data yielded extraordinary, unprecedented predictive capabilities regarding the subject's internal, hidden mental state.³³

Remote Sensing Modality & Machine Learning Algorithm	Predictive Accuracy: Cognitive Stress Detection	Predictive Accuracy: Moral Elevation Detection
r-PPG Only (Support Vector Machine)	77%	61%
Thermal Imaging Only (Random Forest)	79%	78%
Multimodal Fusion (Random Forest)	87%	83%

Table 2: Predictive accuracy of the Multimodal Emotion Recognition (MMER) AI models detailed in the 2024 Liu and Ma study.³³

Through the application of SHapley Additive exPlanations (SHAP) analysis—a method used to interpret the output of complex machine learning models—the researchers proved a critical operational point: while the thermal data was highly accurate on its own, the cardiovascular features covertly extracted via r-PPG exerted the most profound and definitive influence on the AI's ability to conclusively categorize a hidden mental state.³³

7.2. Institutional Alignments in PRC Military-Civil Fusion

The highly advanced research produced by Liu, Ma, and Ni cannot be viewed strictly as benign, theoretical academic inquiry; the institutional affiliations of the authors reveal deep, systemic integration with the PRC's aggressive military-civil fusion apparatus.

The study's primary researchers are affiliated with the Faculty of Psychology at Beijing Normal University and the Shenzhen International Graduate School of Tsinghua University, institutions renowned for their close ties to state security objectives.³³ Most crucially, co-author Fei Ma operates directly out of the Guangdong Laboratory of Artificial Intelligence and Digital

Economy (SZ).³³ This specific laboratory serves as a major, highly strategic nexus for state-sponsored technological research, heavily funded by the National Natural Science Foundation of China and explicitly tasked with advancing the PRC's dominance in the digital sphere.³⁶ The facility's mandate focuses aggressively on advanced machine learning, deep neural networks, cross-modal spatiotemporal feature extraction, and globally scalable cloud computing architectures.³⁶

Furthermore, researchers affiliated with the Guangdong Laboratory actively and regularly collaborate with institutions deeply embedded in the PRC's military-industrial and intelligence complex, including the PLA Information Engineering University, the National University of Defense Technology, and the Defense Innovation Institute at the Academy of Military Sciences.⁴⁰ The laboratory's research output frequently targets explicitly dual-use military applications, encompassing autonomous vehicle motion control and targeting, AI-driven radar micro-Doppler analysis for stealth detection, and cross-domain crowd counting for mass surveillance.³⁹ The publication of the "Your blush gives you away" study, therefore, represents the public acknowledgment of a mature, heavily state-funded capability to remotely, algorithmically, and invisibly profile human psychological states without the subject's consent, awareness, or physical contact.

8. The Cyber-Physical Convergence: Weaponizing the Environment

The geopolitical cyber operations (exemplified by the UNC2814 ICE breach), the deep OT infrastructure vulnerabilities within the electrical grid, the deployment of RF psychoacoustic arrays, and the academic advancements in multimodal biometric AI do not exist as isolated phenomena. Classified intelligence dossiers indicate a terrifying, systemic convergence of these disparate elements into a unified operational theater on Costa Rican soil.³²

8.1. The Uncontrolled Field Laboratory

The core intelligence hypothesis linking these domains posits that state-aligned threat actors, leveraging the deep network penetration established by groups like UNC2814, are actively utilizing compromised residential and corporate environments in Costa Rica as "uncontrolled field laboratories." The primary objective is to test, train, and validate the MMER AI algorithms developed in studies like the 2024 Liu and Ma framework on live, unwitting human targets.³²

In a traditional, sanitized academic setting, inducing cognitive stress requires the voluntary participation of subjects and strictly controlled environmental variables.³³ In the field, the compromised surveillance infrastructure itself provides both the sensory collection apparatus and the psychological stimuli necessary to generate massive volumes of actionable ground-truth data.³²

The myriad physical layer compromises detailed previously facilitate the exact sensory collection required to fuel the r-PPG and thermal fusion AI models ³²:

- **Thermal and Visual ROI Aggregation:** High-resolution cameras, integrated covertly through compromised Israeli-origin Visonic alarm panels and rogue IoT mesh devices (such as the spoofed TP-Link Deco nodes), provide the constant, multi-angle visual stream necessary for the AI to map the 60+ facial Regions of Interest (ROI) required to track thermal vasoconstriction in real time, regardless of the target's location in the environment.³²
- **r-PPG Equivalent Extraction:** The same high-definition visual streams are algorithmically processed on edge devices to detect the micro-fluctuations in skin pigmentation, covertly and continuously extracting the target's Heart Rate Variability (HRV) metrics to gauge sympathetic nervous system activation.³²
- **Wi-Fi CSI Imaging Integration:** The localized network compromises (such as the TR-069 ARRIS backdoor) allow attackers to manipulate Wi-Fi transmission signals.³² Intelligence logs note highly specific patterns of timed deauthentication and disassociation windows.³² These precise anomalies are highly indicative of Channel State Information (CSI) imaging, a sophisticated technique that leverages the phase and amplitude disruption of 2.4 GHz and 5 GHz Wi-Fi radio waves bouncing off human bodies to physically track movement, posture, and even respiration rates through solid walls, providing a constant physical map of the target's location to correlate with the biometric data.³²

8.2. The Ground Truth Feedback Loop

To successfully train an AI model of this complexity, the system must definitively establish a psychological baseline and accurately label the incoming data.³² The threat actors establish this through the implementation of a continuous "Ground Truth Loop".³² By utilizing the constant visual and CSI surveillance, the AI system identifies a baseline physiological trigger—such as a target lighting a cigarette, engaging in a specific posture indicative of work, or engaging in a stressful phone call.³² This organic action provides a highly reliable, easily detectable autonomic nervous system response (such as immediate vasoconstriction and HR alteration from nicotine or stress) that the AI can confidently label as "Baseline Stress".³²

Once the baseline is firmly established, the infrastructure pivots seamlessly from passive biometric collection to active psychological harassment, deliberately injecting stressors to induce cognitive panic and measure the resulting biometric deviations against the baseline.³² The stimuli are delivered continuously via the compromised physical and RF layers ³²:

- **Acoustic and Ultrasonic Harassment:** The localized infrastructure generates piercing ultrasonic bursts with a centroid of ~20 kHz, timed precisely to a pulse repetition frequency (PRF) matching the 46.875 Hz system heartbeat, inducing severe auditory distress.³²
- **Psychoacoustic Entrainment:** The continuous 7 Hz Theta band carrier is utilized to

manipulate the target's baseline cognitive state, eroding critical thinking and rendering them exponentially more susceptible to the subsequent stressors.³²

- **Cyber-Physical Disruption:** Leveraging the unauthenticated Modbus TCP vulnerabilities at the SETECOM OT layer, operators can trigger localized power cycling, flickering lights, or generator anomalies.³² Simultaneously, the TR-069 backdoor facilitates localized, highly targeted network disruptions, instantly severing communications to isolate the target and induce panic.³²

The AI fusion models continuously ingest the target's involuntary physiological responses (r-PPG HRV drops, thermal ROI shifts, respiration changes via CSI) as they react to these orchestrated, multi-sensory disruptions.³² By continuously cross-referencing the applied cyber-physical stimuli against the exact biometric response, the AI system dynamically learns and refines the optimal timing, severity, and frequency of the attacks required to break the target's psychological resilience.³²

The Costa Rican infrastructure, therefore, is not merely being spied upon; it has been fundamentally repurposed into a massive, distributed engine for the refinement of state-level behavioral modification and Neurowarfare technologies. This capability allows the adversaries operating these systems to systematically segment target populations by their psychological susceptibility, effectively mapping the cognitive vulnerabilities of an entire nation.³²

9. Strategic Implications and Conclusions

The cyberattack on the Instituto Costarricense de Electricidad (ICE) by the PRC-nexus group UNC2814 represents a paradigm shift in global cyber warfare. It is not an isolated incident of data theft, but rather a highly visible manifestation of a deeply integrated, multi-domain campaign to achieve total situational and cognitive dominance over a sovereign nation's infrastructure.

The geopolitical catalyst—Costa Rica's strict adherence to the Budapest Convention and the subsequent, courageous exclusion of Chinese 5G vendors—provoked a severe, multi-tiered response from Beijing. The deployment of the GRIDTIDE malware, which brilliantly exploited the ubiquitous Google Sheets API for untraceable Command and Control, demonstrates the highly evasive, cloud-native future of state-sponsored cyberespionage. The exfiltration of 9GB of administrative data provided the adversaries with the macro-level intelligence required to map the state's strategic intentions.

However, the true, existential threat lies beneath the abstracted cloud layer, deeply rooted in the catastrophic vulnerabilities of the operational technology (OT) managed by domestic proxies such as SETECOM. The systemic, institutional failure to secure foundational protocols like Modbus and SNMP at the generator level grants foreign adversaries the latent, terrifying capability to plunge the nation's electrical grid and telecommunications infrastructure into darkness at a moment's notice. This physical compromise provides the ultimate leverage in any

geopolitical standoff.

Furthermore, the convergence of this physical layer compromise with sophisticated Radio Frequency (RF) architectures—encompassing Theta band brainwave entrainment, Frey Effect audio injection, and Extended High Frequency speech extraction—indicates a profound operational shift from passive data collection to active psychological manipulation. The algorithmic foundation for this manipulation is not theoretical; it is actively being pioneered, refined, and published by defense-aligned academic institutions within the PRC, as irrefutably evidenced by the 2024 Liu and Ma study on multimodal emotion recognition. By fusing remote photoplethysmography with thermal imaging, these models allow for the imperceptible, remote tracking of hidden mental states.

The forensic evidence strongly suggests that Costa Rica's compromised networks are actively being utilized as a live field-testing ground. By controlling the cyber-physical environment—manipulating routers via TR-069, tracking human movement through walls via Wi-Fi CSI, and injecting acoustic stressors—adversaries are generating the necessary real-world, ground-truth biometric data required to train the next generation of behavioral modification AI.

The defense against this unprecedented paradigm requires a holistic, uncompromising strategy. Nations must enforce absolute zero-trust architectures at the OT layer, eradicate the use of default credentials in all critical infrastructure, implement deep packet inspection of encrypted cloud API traffic to identify abuses like GRIDTIDE, and physically secure all fiber optic access points against passive interception. Furthermore, intelligence apparatuses must evolve to recognize and counter RF and biometric surveillance techniques. Without these comprehensive measures, the digital and physical infrastructure of sovereign nations will remain a weaponized conduit for inescapable cognitive surveillance and geopolitical coercion.

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